

Personality traits as predictor for individual ambidexterity

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Statement of Originality

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A handwritten signature in blue ink, appearing to read 'M. Poort', with a long horizontal stroke extending to the right.

Menno Poort

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Abstract

This study aimed to examine the influence of personality on individual ambidexterity. Data was collected from 435 employees of a large vocational education institution. Personality was assessed with all the dimensions from the Big Five personality model (TIPI): Extraversion, Agreeableness, Conscientiousness, Openness to experience and Emotional Stability.

Results showed that conscientiousness, openness to experience and emotional stability have a significant relation with individual ambidexterity. Additionally this study shows that dependence among traits can also result in significant relations. Lastly autonomy appeared non-significant as a moderator between the traits and ambidexterity. Implications of these findings are discussed.

Keywords: Personality traits, Exploration, Exploitation, Autonomy, Ambidexterity

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1. Introduction

Agility! Responsivity! Adaptability! Necessities for every firm nowadays to be successful or it seems. But what does it mean and is it something one might learn or are we predispositioned by character? Being adaptive is the ability to switch between exploration and exploitation. The Exploration/Exploitation Balance as Levinthal and March (Levinthal & March, 1993) call it. Organisations need to balance new opportunities to maintain a fit with the future environment whilst maintaining a fitting exploitation of current demands and environment. Running the business today and simultaneously changing tomorrow's business as you will.

This is known as ambidexterity. Previous research has been strongly focused on structural and organisational ambidexterity (Duncan, 1976; Gibson & Birkinshaw, 2004; He & Wong, 2004; Gupta, 2006). Also the individual ambidexterity of managers (Mom, Van Den Bosch & Volberda, 2009; Good & Michel, 2013; Keller & Weibler, 2014; Mom, Fourné & Jansen, 2015; Keller & Weibler, 2015; Bidmon & Boe-Lillegraven, 2019; Tempelaar & Rosenkranz, 2019) has received increasing attention. However, there is little research on individual ambidexterity of the operational level of a firm, i.e. non-managers.

Extant literature has shown that context like structure of an organisation, leadership, coordination and autonomy amongst others influences the ambidextrous behaviour of individuals. Being ambidextrous on an individual level does come with challenges. Individual ambidexterity is more than coupled specialisms and emphasizes on switching back and forth between exploitation and exploration. It has been shown that this leads to cognitive strain and furthermore individuals seem to experience resistance switching to either more exploration or exploitation.

Thus far there is little to no perspective from non-manager personnel in individual ambidexterity literature let alone how their personality effects it. In the research of Mao et al. (2019) a first attempt was made but whilst using an inferior measurement of ambidexterity.

When focussing on the micro-foundations of ambidexterity of an organisation we look at individuals and the context surrounding them. What makes them behave ambidextrous, what inhibits them and what stimulates them. Some individuals might not need stimulation because of an innate disposition instead they should not be troubled by (some) inhibitions. Whilst others need full stimulation to act even slightly ambidextrous. But to assess when and if someone needs stimulation or inhibition first we should know how one's character influences individual ambidexterity.

This research develops and tests hypotheses on the influence of personality traits (IV) on individual ambidexterity (DV) on the operational level of firms (non-managerial personnel). Is there a relation between a person's character or in this case its personality and their subsequent ambidextrous

behaviour. Therefore, the research question is; Are personality traits a predictor of Individual Ambidexterity? Furthermore, I expect that this relation is moderated by the level of (perceived) autonomy of the individual.

To measure ambidexterity, autonomy and to assess the personality several validated constructs are integrated in an online survey. The survey was digitally distributed among the primary target, the teachers of my organisation (approximately 1035 colleagues). From the 1035 recipients of the invitation to take part in this research 495 respondents started filling out the survey, 435 respondents fully completed the questionnaire (response rate 42%).

The direct effects of single traits on individual ambidexterity was hypothesized and tested through regression analysis. Of the five traits conscientiousness, emotional stability and openness to experiences showed significant relations to individual ambidexterity. Moreover the (inter)dependence of traits was hypothesized and tested though the effect of the latter traits on extraversion. This too resulted in significant results were extraversion (with conscientiousness low or emotional stability high) has a positive effect on individual ambidexterity. Finally autonomy was hypothesized as an moderator on the relation between the traits and individual ambidexterity. This was tested as a general effect on all traits at once and on each individual traits with high and low levels of autonomy. No significant effect was found.

In extension to the title of Barrick's research (2005); Yes, personality matters... in relation to individual ambidexterity. This study brings us a small step closer to fully understanding individual ambidexterity were personality might be the starting point in creating an optimal context to facilitate ambidextrous behaviour.

2. Literature review

2.1. Ambidexterity

Before elaborating on individual ambidexterity which is the main focus of this study a short introduction of ambidexterity is in order. Balancing both exploration and exploitation is often referred to as ambidexterity. In the past authors have referred to ambidexterity in structural terms. According to Robert B. Duncan (1976), who introduced the term, 'organizations achieve trade-offs between conflicting demands by creating "dual structures," in a way that some business units or groups focus on alignment, while others focus on adaptation' (Duncan, 1976).

Exploration is described by several scholars as; "Searching for new possibilities, evaluating diverse options, adaptability, new skills" (Mom et al., 2009) and "Behavior that includes search for alternatives and disengagement from the current task" (Laureiro-Martínez, Brusoni & Zollo, 2010). The same goes for exploitation; "A lot of experience, routine, short-term goals, present knowledge" Mom et al (2009) and "Engaged in behaviours associated with the most valuable (states) that it has already discovered" (Aston-Jones & Cohen, 2005).

To create consistency throughout this research I will refer to exploration and exploitation instead of adaptation and alignment (Gibson & Birkinshaw, 2004), (Duncan, 1976) and therefore use March' (March, 1991) definition of exploration and exploitation; 'Exploration includes things captured by terms such as search, variation, risk taking, experimentation, play, flexibility, discovery, innovation. Exploitation includes such things as refinement, choice, production, efficiency, selection, implementation, execution.'

These exploratory and exploitative activities need to be mobilized, coordinated, integrated, and applied. Gupta (2006) refers to ambidexterity as 'the synchronous pursuit of both exploration and exploitation via loosely coupled and differentiated subunits or individuals, each of which specializes in either exploration or exploitation.'

So now that a clear understanding of the (underlying) principles of ambidexterity is established, it cannot be ignored that all of these descriptions can be applied at an organisational, subunit or interpersonal level. This could be interpreted as a predisposition to either exploration or exploitation which leads to ambidexterity at a higher organisational level. So what about the intrapersonal level of analysis, is ambidexterity possible on an individual level and what are the predispositions?

2.2. Individual ambidexterity

This study focus' on individual ambidexterity which is described as; 'the individual-level cognitive ability to flexibly adapt within a dynamic context by appropriately shifting between exploration and exploitation' (Good & Michel, 2013). Or as the "Cognitive flexibility to recognize the advantages and disadvantages of the two alternative allocations (exploration and exploitation)" (Laureiro-Martínez et al., 2010). So the latter descriptions elaborate on that of Gupta (2006) in the sense that individual ambidexterity is more than coupled specialisms and emphasizes on switching back and forth between exploitation and exploration. Most importantly both definitions focus on the individual level.

The research of Tempelaar (Tempelaar & Rosenkranz, 2019) focuses on the microfoundations of organizational ambidexterity. Trying to understand what enables individuals to address the exploitation-exploration dilemma. Ambidexterity challenges individuals through 'the multiplicity and divergence of organizational roles'. Individuals need to accommodate this multiplicity and divergence when exploiting and exploring. Tempelaar et al (2019) specifically point to the importance of how individuals identify with and enact this multiplicity of role demands.

They find that some persons may show ambidextrous behaviour through organisational means (for example cross-functional team) while others are more intrinsic oriented to act ambidextrous. By focusing on personality in this study the aim is to further zoom in on this intrinsic orientation and if there is a relation at all.

Bidmon and Boe-Lillegraven (2019) further explore Individual ambidexterity as an important micro-foundation of organizational ambidexterity. In line with Tempelaar et al (2019) they state that switching back and forth between exploration and exploitation can be challenging for individuals. In contrast to prior research that mostly focused on bottom-up approaches to stimulate individual ambidexterity, Bidmon et al (2019) concentrates on top-down control since many organisations are characterized by the latter rather than the former. Their research confirmed a tendency of individuals to resist switching by reinforcing either exploration or exploitation.

A solution could be to stimulate an individuals' ambidextrous behaviour by structural control (reduction in autonomy) and contextual support where expectations to switch back and forth between exploration and exploitation are accentuated. Therefore, specific conditions (autonomy, task design and formats) surrounding an individual which might be crucial in connecting organisational ambidexterity with individual ambidexterity. However it does not connect the identity or personality of an individual to ambidexterity.

Keller and Weibler (2015) contribute to the developing understanding of ambidexterity at the individual level by studying how leadership is related to individual ambidexterity and how this is related to cognitive strain. Their research shows that a manager's level of ambidexterity to be positively related to cognitive strain. This relationship is positively moderated by the level of conscientiousness and negatively moderated by the level of openness to experience. Even though the level of analysis is that of managers, the reasoning of cognitive strain and the effect of traits on ambidexterity is already on an individual level, but is it distinct for managers or does it account for non-managerial personnel as well? The higher the level of ambidexterity the higher the cognitive strain were conscientiousness relieves the strain and openness to experience increases it. So some traits increase the ability to behave ambidextrous and some decrease that same ability. This study aims to further contribute to the understanding of this apparent mechanism.

2.3. Individual predictors: Personality

Mao and Washida (2019) found that for research staff at universities there is a relation between the personality trait openness to experiences and researchers' ambidextrous behaviour as well as their exploratory behaviour. While Keller and Weibler (2014) did connect personality to explorative or exploitative behaviour they did not connect personality to ambidextrous behaviour like Mao et al. (2019) did. Therefore a relation between certain traits and ambidexterity apparently exist but are they all related and in what way? And is it also applicable to other sectors? First a closer look in what personality traits are and if they matter.

To start, we all have personalities. Everybody has all traits in different levels (Barrick, 2005). Furthermore there is prevalent consensus that noncognitive factors (like personality traits) are strongly associated in many or perhaps most aspects of job-related performance. To put it more strongly; "Intellectually able individuals falter on the job when their personality traits are not congruent with task requirements" (Goldberg, 1993).

Research by Barrick (2005) used seven divergent research streams to substantiate on empirical grounds that personality predicts job performance across a wide variety of outcomes that organizations value. So in line with the title of Barrick's (2005) research; Yes, Personality Matters (Moving on to More Important Matters).

But which traits are there and what do they express? Emotional Stability (other side of the spectrum is Neurosism) and Conscientiousness, generalize in the prediction of overall performance and refer to the ability to assign resources to complete tasks (Emotional Stability), and the inclination to follow rules and to apply effort (Conscientiousness). These predictors can be best regarded as measures of trait-oriented work motivation. Moreover it appears they affect performance in all jobs (Barrick,

2005). To influence the ability to adapt to change and creativity have been related to Openness to Experience (George & Zhou, 2001). Additionally Openness to experience is associated to curiosity, attitudinal and behavioural flexibility, creativity, intellect, and divergent thinking (George & Zhou, 2001) (Barrick & Mount, 1991). The traits conscientiousness and openness to experience are related to facets of learning behaviour (Barrick & Mount, 1991). The traits Openness to Experience, Agreeableness and Extraversion are also valid predictors of performance. But, according to Barrick (Barrick, Mount & Judge, 2001) only in precise positions, for specific jobs or for some criteria. Agreeableness seem to matter merely when interaction involves cooperating, aiding and caring for others. In line with the latter it is safe to say that if participating in a team is a predominant factor, Agreeableness may be the single best personality predictor (Mount, Barrick & Stewart, 1998). Extraversion seem to be related to job performance in professions where a substantial part of the job involves interacting with others, specifically when the interaction is concentrated on influencing others and obtaining power and status (Barrick et al., 2001).

Interdependence

Further research (Witt, 2002) found that conscientiousness was unrelated to performance among workers who score very low on extraversion. Extraversion seemed positively related to job performance among the highly conscientious workers but negatively related among low-conscientiousness workers.

Low-conscientious and low-extroverts as well as high-conscientious and low-extroverts established lower performance scores than the high-conscientious high-extroverts. To put it differently, without extraversion, conscientiousness may enhance performance insufficient in jobs demanding interpersonal interaction. However low-conscientious high-extroverts score the lowest performance ratings. High-extroverts with low-conscientiousness may yield dysfunctional behaviour in jobs demanding interpersonal interaction.

As mentioned above there is extant research about personality and the specific traits, their interdependence and especially in relation to job performance or organisational fit, but very little is focused on the relation of personality to ambidexterity.

As introduced earlier I am interested in a possible relation between a person's personality and individual ambidexterity. Mao et al. (2019) found a relationship but there are factors that have to be taken into account. First, their target group were researchers of universities and private enterprises almost all with an education level of bachelor or higher. In my research, I will use a broader educational perspective and subsequently a more heterogeneous target group. Furthermore their research state that the quality of the model can be considered due to low quality of question items

on measurement of researchers' ambidexterity. Therefore, I will use a validated method of measuring ambidexterity (Mom et al., 2007; Keller & Weibler, 2014).

2.4. Contextual predictor: Autonomy

Regarding ambidexterity research often the contextual predictor, whether moderator or mediator, was in fact ambidexterity itself. Ambidexterity has been used as mediator by Gibson and Birkinshaw (2004) in relation to performance (effect was significant ($z=3.22$)). And also Mao et al. (2019) used ambidexterity as mediator between traits and journal publications and research expenses but resulted in only a significant effect between traits and ambidexterity and no significant effect between ambidexterity and the two DV's.

When looking at the individual level of ambidexterity in research this changed. Keller and Weibler (2015) use openness to experience and conscientiousness as moderator between ambidextrous balance and cognitive strain ($\beta=-.18$, $p<.05$; $\beta=.17$, $p<.05$). Additionally Mom et al. (2019) used Role-breadth self-efficacy and Intrinsic motivational orientation as mediators between firm HR practices and operational manager ambidexterity. (RBSE; indirect effect = 0.15, $p = .000$, 95% CI [0.10, 0.20], IMO; indirect effect = 0.12, $p = .000$, 95% CI [0.07, 0.16]). (Mom et al., 2019).

These all relate to internal factors of individuals (predominately managers) and if they influence ambidexterity. Others relate to more external factors of individuals such as leader behaviour (Ng et al., 2008), functional tenure (Mom et al., 2015) and Tempelaar and Rosenkranz (2019) use cross-functional coordination as moderator between role transition and individual ambidexterity ($\beta=.35$, $p<.001$). And lastly the degree of control and autonomy granted (Bidmon et al., 2019). The latter is interesting since extant literature has been written about autonomy and the effect on (job) performance (Gellatly & Irving, 2001) (Langfred, 2005) (Lanaj et al., 2013)) but not on ambidexterity. Bidmon and Boe-Lillegraven (2019) use scripts in their research to direct top to bottom. These scripts consist, amongst others, clear instructions and in that sense reduce autonomy to support and enable switching behaviour necessary for individual ambidexterity. This suggests that less autonomy might help in activating individual ambidexterity. Since their research was conducted through observations of 38 switching situations the generalization of their findings might ask for further examination of autonomy and the effect on individual ambidexterity.

In this study the variable autonomy is exemplified as job autonomy which refers; "to the degree of latitude employees possess in making job-related decisions, such as what tasks to perform, how the work is to be done, and how work exceptions are to be handled" (Hackman & Oldham, 1975). Based

on the latter research a measurement of work autonomy was constructed (Breugh, 1985) which will be explained in chapter 3. Job autonomy has been theoretically linked to job performance (Hackman, & Oldham, 1975) but since the effect of autonomy is still underdeveloped in individual ambidexterity research it will be taken into account in this study.

2.5. Hypotheses and research model

This research aims to prove that personality traits can be a predictor of individual ambidexterity. This might be moderated by the level of autonomy. Therefore, the research question is; Are personality traits a predictor of Individual Ambidexterity?

Barrick (2005) stated; "When the purpose is to enhance understanding of which personality constructs predict which components of performance, it is appropriate to examine the validity of each personality trait individually. However, when the purpose is to maximize prediction, the appropriate approach is to consider the validity of all relevant personality traits when used together." Therefore in this study will aim at the understanding of individual traits and the interdependence of some traits and the relation with ambidexterity.

Are there certain personality traits that can act as a predictor of individual ambidexterity? Mao et al (2019) showed there Openness to experiences (.485, $p = .001$), Conscientiousness (-.13 $p = .05$) are significantly associated to ambidextrous orientation. Extraversion, Emotional stability (opposite of Neuroticism) and Agreeableness showed no significant relationship.

As stated before Keller and Weibler (2015) found that the higher the level of ambidexterity the higher the cognitive strain were conscientiousness relieves the strain and openness to experience increases it. So some traits increase the ability to behave ambidextrous and some decrease that same ability. This research aims to further contribute to the understanding of this apparent mechanism between traits and ambidexterity therefor the following hypotheses will be tested;

H1a: Extraversion has a significant effect on Individual ambidexterity

H1b: Agreeableness has a significant effect on Individual ambidexterity

H1c: Consciousness has a significant effect on Individual ambidexterity

H1d: Emotional Stability has a significant effect on Individual ambidexterity

H1e: Openness to Experiences has a significant effect on Individual ambidexterity

Additionally are there constellations of high or low level traits that have a significant effect? The research by Witt (2002) showed an interactive effect of conscientiousness and extraversion in

relation to job performance. Specifically that extraversion had a positive relation to job performance among the highly conscientious employees but negatively related among low-conscientiousness employees.

Thus could this also account for individual ambidexterity? In Mao et al (2019), Witt (2002) and Keller and Weibler (2015) conscientiousness plays a significant role. So this might suggest that one needs a high level to behave ambidextrous. The same goes for openness to experience although it increases cognitive strain (Keller, T., & Weibler, J., 2015).

Furthermore the cognitive strain associated with ambidexterity might be decreased by emotional stability in the sense that remaining calm and not losing composure has a positive effect when accommodating the multiplicity and divergence of exploration and exploitation (Tempelaar & Rosenkranz, 2019). In line with Witt (2002) extraversion will be tested to investigate interactive effects (on ambidexterity instead of performance) with the following hypotheses;

H1x: The relationship of extraversion with ambidexterity is dependent upon the level of conscientiousness.

H1y: The relationship of extraversion with ambidexterity is dependent upon the level of openness to experience.

H1z: The relationship of extraversion with ambidexterity is dependent upon the level of emotional stability.

Lastly the possible relationship of traits with ambidexterity might be dependent on the level of autonomy a person has. This was observed by scholars of work design literature that autonomy (decentralized planning amongst others) has positive effects for flexibility and creativity (related to exploration) but also negative effects for efficiency and coordination (related to exploitation) (Lanaj et al., 2013).

This seems contradictory since autonomy affects both kind of activities related to ambidexterity but in opposing ways. By increasing autonomy, exploration goes up but exploitation goes down, thus logically ambidexterity might be difficult to achieve. Although it was not tested on ambidexterity itself let alone on personality traits.

Mao et al (2019) showed that openness to experience has a positive effect on both exploration and exploitation eventually resulting in a positive effect on ambidexterity. So perhaps can autonomy strengthen this effect because openness to experience is beneficial on all sides. But since some traits might have a positive and others a negative effect on either exploration, exploitation or

ambidexterity it might be that some trait/ambidexterity relations are strengthened and others weakened by autonomy.

Hence, is the relation of certain traits with ambidexterity then dependent on the level of autonomy? And does it affect all in the same direction? Someone who has a low level of conscientiousness might not benefit from high level of autonomy, less structure and control is not a good thing for a unstructured person. So different kind of people are suited for different kind of situations (Tempelaar & Rosenkranz, 2019).

In research regarding job autonomy (Ng et al., 2008) results showed that the conditional indirect effects for conscientiousness and neuroticism (Emotional stability) were significant in the high job autonomy condition (neuroticism = $-.072$, $p = .01$; conscientiousness = $.092$, $p = .01$) but were not significant under low job autonomy (conscientiousness $.019$, ns, neuroticism = $-.019$, ns). They found that job autonomy will moderate the strength of the mediated relationships between leaders' neuroticism and conscientiousness.

Thus high autonomy does seem to have effect. Further results in the same research also show that there was no moderated mediation for extraversion at both high and low job autonomy, they were significant but did not differ much from each other (high $.042$, $p .05$, and low $.071$, $p .05$).

When combining the latter three paragraphs it brings the idea that specific traits might be affected differently by autonomy and that the relation with ambidexterity might also be dependent on the level of autonomy.

This leads to the following hypotheses;

H2a: The relation of Extraversion with individual ambidexterity is negatively moderated by higher rather than lower levels of individual autonomy

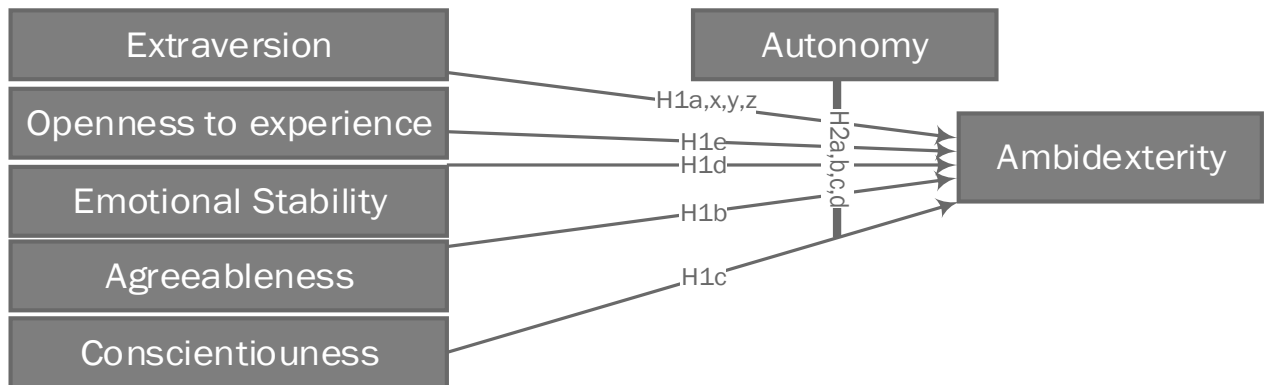
H2b: The relation of Conscientiousness with individual ambidexterity is positively moderated by higher rather than lower levels of individual autonomy

H2c: The relation of Emotional stability with individual ambidexterity is negatively moderated by lower rather than higher levels of individual autonomy

H2d: The relation of Openness to experience with individual ambidexterity is positively moderated by higher rather than lower levels of individual autonomy

The schematic depiction of the research design can be seen below in the Conceptual model.

Figure 1: Conceptual model



3. Method

The chapter Method represents the start of the empirical part of this research. Firstly the most apparent characteristics of the collected sample will be outlined. Subsequently, the variables included in the questionnaire and corresponding reliabilities are discussed. A brief description will be given of the statistical approach that was taken in order to test for the expected hypotheses in the previous chapter. See the appendix for the complete questionnaire (in Dutch).

3.1. Sample

The target group of this research consists of in theory all operational level personnel, but in practise, the sample consisted of all the teachers within ROC Midden Nederland. ROC Midden Nederland is a large organisation of vocational education with over 1600 employees and 20.000 students. Since vocational education consists of teaching most of the core crafts in every sector (from nurse to car mechanic to hair dresser) most teachers need to balance structured education plans and external trends and influences of their specific sectors. Moreover is the educational sector known for high levels of structure and control so apparently this results in low levels of individual autonomy.

The survey (see appendix 7.1) was digitally distributed among the primary target, the teachers of my organisation (approximately 1035 colleagues). Secondly, the survey was distributed through my network, LinkedIn specifically, to assess if internal results match external results as a validity measure. External validation resulted in 24 respondents which is a too small sample to substantiate possible results from the internal survey therefore it will be excluded from this research.

From the 1035 recipients of the invitation to take part in this research 495 respondents started filling out the survey, 435 respondents fully completed the questionnaire (response rate 42%).

Of these 435 respondents together ($M_{age} = 48.1$, $SD_{age} = 12.0$, age-range: 23-66 years) 53.8% were female. Furthermore the majority of 63.7% completed an educational programme at bachelor level (HBO and University). 6% had a high school or vocational educational level degree (VMBO, Havo, VWO, MBO). The remaining 30,3% had a master's degree (HBO and University).

Of the respondents 0.5% reported an organizational tenure ($M_{tenure} = 12.5$, $SD_{tenure} = 10.6$, tenure-range: 0-42) less than a year, 39.8 % reported an organizational tenure of 1-5 years, 14.5% reported an organizational tenure of 6-10 years, 9.9% reported an organizational tenure of 11-15 years, 14.0% reported an organizational tenure of 16-20 years, 6.9% an organizational tenure 21-25 years, 6.7% reported an organizational tenure of 26-30, 5.3% reported an organizational tenure of 31-35 years and 2.5% reported an organizational tenure of more than 35 years. The majority of the respondents reported a job tenure ($M_{jobtenure} = 10.5$, $SD_{jobtenure} = 9.7$, job tenure range: 0-41) of less than 10 years (65.5%). Within this group 44.8% had a job tenure of less than 5 years.

3.2. Measurement of variables

Translation, back-translation procedure

Most items used in the questionnaire were derived from English studies. Because all respondents to the survey had Dutch as their first language, the original items were translated into Dutch. To assure that the content of the items remains unchanged, the translated Dutch items were back translated into English by a third person. Furthermore a test survey was distributed among five persons to check the translation and comprehension of specific constructs (Exploration and Exploitation (Keller & Weibler, 2015)) and the flow of the survey. The check on translation and comprehension resulted in slight adjustments in phrasing but remained very close to the original.

The Big 5 personality test

The Big 5 personality test usually consist of a multitude of questions. This would render a survey too time consuming if using a classic Big 5 questionnaire. In this research a Ten-Item Personality Inventory (TIPI) (Gosling, 2003) was applied. Even though this instrument is slightly inferior to standard multi-item instruments, it reached adequate levels in terms of convergence with widely used Big 5. The findings in the research of Gosling (2003) suggest that TIPI can stand “as reasonable proxies”. The TIPI is constructed through ten statements with a seven item Likert scale. An example item is “I see myself as Extraverted, enthusiastic.”

To test if some traits have a dependence of other traits the analysis will be conducted with high (+1 SD), average (0), and low (–1 SD) levels of expected dependent traits (H1x, y and z) (Cohen, J., & Cohen, P., 1983).

Ambidexterity

To measure ambidexterity, Keller and Weibler (2015) used a 16-question construct where both exploitation and exploration factors are loaded on to by 8 questions each. This research is based on that of Mom et al (2007).

This model is constructed through 16 questions with a seven item Likert scale. An example item is “To what extent did you, last year, engage in work related activities that can be characterized as follows: Activities in which you have to deal with previously unknown situations.” Even though the constructs of Mom et al. (2007) and Keller and Weibler (2015) are related they differ in questions and phrasing. An example is; “To what extent did you, last year, engage in work related activities that can be characterized as follows: Searching for new possibilities with respect to products/services, processes or markets” (Mom et al., 2007).

Both are items which load onto Exploration, but the phrasing of Mom et al (2007) has a more

corporate ring to it than the ones of Keller et al. (2015). During preliminary testing of the survey (as mentioned above) the feedback was to keep the phrasing as simple as possible and to reframe from corporate wording to make it as assessable as possible. To put it simple, keep it simple stupid, was the adage. Therefore the construct of Keller and Weibler (2015) was used with the adjustment of remembering a week instead of a year. The latter was on the premise that a respondent can more easily remember their activities of a normal working week than of an entire year.

Keller and Weibler (2015) used the absolute difference between exploration and exploitation to operationalize ambidexterity as an balance of both patterns in managers' behaviour, as this is consistent with He and Wong (2004) and Cao et al. (2009). Secondly they subtracted the absolute difference from five, so that a higher ambidexterity score represents a closer balance between exploration and exploitation in a manager's behaviour.

Autonomy

As a moderator the task autonomy (perceived or experienced) of operational level personnel was included in this research. To assess autonomy, the survey consists of the 9 item framework by (Langfred, 2005) which is based on the work of Breugh (1985) in 'The Measurement of Work Autonomy', a well-validated and frequently used scale for the measurement of individual job autonomy (reliability Cronbach's alpha of .90). An example item is "I am free to decide how to go about getting my work done."

To test if autonomy has an U-shape regression analysis will be conducted with high (+1 SD), average (0), and low (−1 SD) levels of autonomy (Cohen & Cohen, 1983).

Control variables

Furthermore the study will be controlled by the following six control variables; Gender (since the sample captured both female and male respondents) Age, Education, Tenure in organisation, Tenure in current job and Sector (internally related to colleges). Age, Education and Tenure is used to control for experience in line with Keller and Weibler (2015) and Mom et al (2009).

Regarding (organizational) tenure Mom et al (2015) predicted a positive relationship with (a managers) ambidexterity. The coefficient for organizational tenure is positive and significant ($\beta = 0.25$, $p < 0.001$) related to ambidexterity (Mom et al., 2015).

3.3. Statistical

Data collection started via Qualtrics on April 14th 2020 after respondents received a personalised e-mail inviting them to participate and priming them to take into account a normal work week. Priming them was necessary due to the outbreak of COVID-19 and the official lockdown in the Netherlands

making a regular working week at that time everything but normal. To ensure the most effective priming possible the respondents were reminded to take a normal work week into account in the introduction of the survey itself. The survey was closed on April 25th 2020. Through the Statistical software Package for Social Sciences (SPSS 25) the statistical analysis was performed.

Missing values

In total the survey had 494 respondents but 53 did not finish and were therefore removed by listwise deletion. Further inspection of all variables resulted in 2 missing values (Q6) and 3 invalid results (Q9), all of which were removed by listwise deletion. The main constructs of the survey were complete and valid. The last 2 questions both had missing values but they did not affect the main constructs so those were ignored. To draw conclusions for a great variety of operational level personnel behaviours and to strengthen the external validity of the results, the hypotheses were evaluated using a sample of 435.

Recoding

The construct of the independent variable consists of 10 questions were pairs of 2 create a value of a personality trait. In all of the 5 traits there was a reversed scored question that was recoded (Q1_2, Q1_4, Q1_6, Q1_8, Q1_10).

Normality

After recoding, the scale means were computed for the traits and the constructs of Exploration, Exploitation and Autonomy. And Ambidexterity was computed using Exploration and Exploitation (see 3.2). Furthermore descriptive statistics, skewness, kurtosis and normality tests were computed. Of the 14 variables, only Ambidexterity was normally distributed. Tenure (job and organisational) had a flatter and more positive distribution.

Most of the traits were moderately negatively skewed (except for Agreeableness (.403)) and pointy (except for Extraversion which had a negative kurtosis (-.272)). Exploration and Exploitation both had a moderate negative skewness (-.371 and -.707) were the latter had a more pointy distribution due to a positive kurtosis (.406) compared to the first (-.360). Autonomy was negatively skewed (-.924) and had a positive kurtosis (1.438).

Even so skewness will not make a fundamental difference in the analysis due to the large sample size (Tabachnick & Fidell, 2001). Furthermore “Kurtosis can result in an underestimate of the variance, but this risk is also reduced with a large sample” (Tabachnick & Fidell, 2001). In this research there

were more than 400 respondents, therefore the risk is reduced and both skewness and kurtosis should not make a substantive difference in the analysis.

Reliability

The reliability checks were run on all main constructs related to the hypothesis, personality traits (TIPI), exploration, exploitation and autonomy.

Exhibited in table 1, three main variables (exploration, exploitation and autonomy) have a Cronbach's alpha > .80, which indicates high level of internal consistency. With all the three variables the corrected item-total correlations indicated that all the underlying items have a good correlation with the total score of scale (all above .30 and most even above .50). Furthermore, none of the items would substantially affect reliability if they were deleted.

Table 1: Reliability

VARIABLE	CRONBACH'S ALPHA	CRONBACH'S ALPHA ORIGINAL RESEARCH
PERSONALITY TRAITS (TIPI) OVERALL	,428	
EXTRAVERSION	,693	,68
AGREEABLENESS	-,129	,40
CONSCIENTIOUSNESS	,473	,50
EMOTIONAL STABILITY	,523	,73
OPENNESS TO EXPERIENCES	,416	,45
EXPLORATION	,845	,83
EXPLOITATION	,841	,83
AUTONOMY	,859	,90

The variable Personality (TIPI) has an overall alpha of ,428 which means a poor reliability (< .50). This is not a problem since the overall construct is not used. But the specific traits are. But due to the fact that every trait consists of only two items per scale it is expected that they result in lower inter-item correlations than is archetypal of more homogenous scales.

The TIPI is a less reliable construct than for example a 44 item Big-Five Inventory (John & Srivastava, 1999). Furthermore single-item measures cannot assess or control for acquiescence bias (tendency of respondents to select the positive response option) (Paulhus & Vazire, 2007) and in this case it is expected that controlling for this bias is also difficult for two-item measures.

This might explain the strong contrast of the observed alpha of agreeableness versus that of the original research. The same might explain the difference of emotional stability. Since this research is

based on a one time survey test-retest reliability is not an option. Due to the negative alpha of agreeableness this variable is dropped.

Computing scale means

Concluding the preliminary steps before further analysis the scale means was compute of the variables; exploration, exploitation and autonomy. All three variables (Q4, Q5 and Q11) consisted of 8 or 9 items. And to compute the personality traits as mentioned before all the five traits consisted of the means of two items were one of them was recoded (Q1_2, Q1_4, Q1_6, Q1_8, Q1_10).

Principal component analysis

The constructs exploration, exploitation and autonomy were checked for characteristics, dimensionality (principal component analysis with varimax rotation), and reliability (Cronbach's alpha). To confirm that the scales of the constructs are appropriate for measuring exploration, exploitation and autonomy in this study, confirmatory factor analyses via SPSS was conducted.

Factor loadings of all the items are at least .50 and meet the criteria (Bagozzi & Yi, 1988), except for one autonomy item. However, this item was not excluded because it is considered to capture an important aspect of autonomy (items are given in the appendix 7.2).

Computing Ambidexterity

Keller and Weibler (2015) computed the absolute difference between exploration and exploitation (see 3.2 Measurement for construction of both variables). Subsequently the absolute difference between exploration and exploitation was subtracted from five, so that a higher ambidexterity score represents a closer balance between exploration and exploitation in behaviour.

4. Results

Firstly a short summary of the descriptive statistics is given. Table 2 shows the descriptive statistics and correlations. As stated earlier (3.1. Sample) the total sample size consisted of 435 respondents.

The majority is female (53.8 %) and the average age is 48,1 years. The average tenure is 10,4 years in relation to current job and 12,4 in relation to organisational tenure. All the traits have a mean above 5 which indicates that on average the sample scores slightly higher since they are measured on a seven-point Likert scale. The same scale is used for exploration, exploitation and autonomy. In that sense exploration with a mean of 3,9 scores lower than exploitation with a mean of 5.2. So on average the sample is more inclined to exportation than exploration. Autonomy has a mean of 5 which also indicates that on average the sample score their autonomy as moderately high. Ambidexterity has a mean of 6.3 which is a relatively high score (considering the method described on page 22) and reflects the higher mean of exploitation in contrast of that of exploration.

4.1. Correlation analysis

Since this research aims on finding a predictor of ambidexterity through personality traits (and their possible interdependence) table 2 contains the independent and control variables. Control variables (Gender, Age and Education) show no correlation with ambidexterity.

The traits Conscientiousness ($r=,111$, $p=0.05$), Emotional stability($r=,141$, $p<0.01$) and Openness for experiences ($r= -,132$, $p<0.01$) have a significant correlation to ambidexterity (8). The traits also show correlation amongst themselves, Openness to experience is positively correlated to Extraversion but negatively with Conscientiousness. Emotional stability is positively correlated to Conscientiousness and Conscientiousness is negatively correlated to Extraversion.

Furthermore both organisational and job tenure are correlated to age (Mom et al, 2019).

Table 2: Descriptive Statistics and Correlations.

Variable	Items	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1 Gender	1	1,462	0,499													
2 Age	1	48,143	12,015	,098*												
3 Education	1	4,237	0,573	-0,09	,116*											
4 Job Tenure	1	10,483	9,660	0,092	,515**	0,045										
5 Organisational Tenure	1	12,487	10,584	0,035	,597**	,127**	,633**									
6 Extraversion	2	5,331	1,344	-,105*	-0,06	-0,05	-0,06	-0,09	(,693)							
7 Conscientiousness	2	5,991	0,904	-,197**	0,089	0,004	0,082	,105*	-,101*	(,473)						
8 Emotional Stability	2	5,595	1,049	,149**	,102*	-0,01	0,034	-0,03	-0,04	,146**	(,523)					
9 Openness to Experiences	2	5,483	1,012	-0,05	-0,03	-0,02	-,197**	-,220**	,262**	0,029	,158**	(,416)				
10 Exploration	8	3,907	1,092	0,01	0,08	0,042	-0,01	0,055	-0,02	-,108*	-0,09	0,061	(,845)			
11 Exploitation	8	5,210	0,903	-0,01	0,003	-0,02	,126**	0,071	0,021	0,066	,137**	-,160**	-,271**	(,841)		
12 Ambidexterity	2	6,303	1,595	-0,01	-0,05	-0,04	0,075	0,002	0,024	,111*	,141**	-,132**	-,839**	,752**		
13 Autonomy	9	5,020	0,933	0,045	-,164**	0,003	-,184**	-,144**	,133**	-0,08	,103*	-0,03	0,008	,139**	0,073	(,859)

Note: N = 435. Reliabilities are reported along the diagonal.

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

4.2. Direct effects

A hierarchical regression analysis is used to analyse if the personality traits (the independent variables) have a predictive power for all dependent variables and Ambidexterity.

Ambidexterity

As shown in table 2 significant correlations between three personality traits and Ambidexterity imply direct relationships. In order to examine the linear relation between the independent variables, Extraversion, Agreeableness, Conscientiousness, Emo_Stability and Open_Experiences, and the dependent variable Ambidexterity a second regression analyses was conducted, see table 3. As a first step three predictors were entered, the control variables: gender, age and education as model 1. Model 1 was statistically insignificant $F(3, 431) = .570$; $p > .05 = .635$ and therefor explained 0% variance in Ambidexterity. Although the population displayed large overall variations in gender, age and education level, it was not related to Ambidexterity. So most of the variance of the variable Ambidexterity is explained by other factors. In Model 2 the five personality traits were entered both together and independently as a predictor. Together the total variance explained by the model as a whole was 7% $F(8, 426) = 4,005$; $p < .001$.

The independent analysis of the traits show that three out of five predictor variables were statistically significant, with Conscientiousness showing a slightly lower Beta value ($\beta = .119$, $p < .05$) than Emo_Stability ($\beta = .151$, $p < .01$). Openness to Experiences has a negative Beta value ($\beta = -.136$, $p < .01$). Even though these traits are significant they explain only a small portion of the variance of the variable Ambidexterity. Emotional Stability explains 2,6%, Openness to Experiences explains 2,2% and Conscientiousness 1,7%. All traits were tested on multicollinearity through collinearity statistics whilst performing the regression analysis. All the four traits had a tolerance around 0.9 and a VIF around 1,1 therefore no collinearity detected.

Table 3: Results of Extraversion, Conscientiousness, Emotional stability and Openness to experience as predictors of individual ambidexterity

Dependent Variable: Ambidexterity	R	R Square	R Square Change	B	Std. Error	Beta	t
Model 1	,063a	0,004	0,004				
Gender				-0,038	0,155	-0,012	-0,244
Age				-0,006	0,006	-0,048	-0,990
Education level				-0,091	0,135	-0,033	-0,675
Model 2	,264b	0,070***	0,066***				
Extraversion	,066	,004	,000	,022	,058	,019	,386
Conscientiousness	,132	,017*	,013*	,210	,087	,119*	2,426
Emo_Stability	,162	,026*	,022*	,230	,073	,151**	3,132
Open_Experiences	,149	,022*	,018*	-,214	,075	-,136**	-2,845

a. Predictors: (Constant), Education, Gender, Age

b. Predictors: (Constant), Education, Gender, Age, Open_Experiences, Agreeableness, Conscientiousness, Extraversion, Emo_Stability

Statistical significance: * $p < .05$; ** $p < .01$; *** $p < .001$

So if the level of Conscientiousness increases for one, the level of Ambidexterity raises 0,119. The same goes for Emotional Stability were if the level increases by one the level of Ambidexterity increases by 0,151. The counter effect is seen with Openness to Experiences, if the level increases with one the level of Ambidexterity decreases with 0,136.

Due to the fact that Extraversion has no significant effect on Ambidexterity, Hypothesis 1a is rejected. Hypothesis 1b is discarded due to the fact that Agreeableness was excluded because of low reliability. Concluding Hypotheses 1c, d and e are supported since these three traits have a significant effect on Ambidexterity.

Interdependent effect of traits

To further test the hypotheses H1x, y and z regression analysis was conducted with the interdependent trait filtered on high (+1SD) and low (-1SD) levels. So to test H1z conscientiousness was filtered ($\geq 6,895$ (mean + sd)) and then the analysis was run. To act as an additional validity check the same analysis was conducted using the z-variables of extraversion and conscientiousness (Witt, L. A., 2002). This check resulted in the same values as the non z variable analysis. All traits were tested on multicollinearity through collinearity statistics whilst performing the regression analysis. All the four traits had a tolerance around 0.9 and a VIF around 1,1 therefore no collinearity detected.

Table 4: Interdependent effect of traits on individual ambidexterity

Regression interdependence of traits with Ambidexterity						
Predictors	β					
	Conscientiousness		Openness to experience		Emotional stability	
	High	Low	High	Low	High	Low
Step 1						
Gender	,252*	-,173	-,111	-,005	,133	,068
Age	-,198	-,042	-,109	,15	-,155	-,031
Education	,052	,097	-,121	-,25	,06	-,083
R2	,078	,043	,04	,084	-,068	,015
Adjusted R2	,049	,009	,012	,028	-,039	-,018
Step 2						
Extraversion	,056	,251*	-,068	-,216	,38*	,118
R2	,081	,103*	,045	,131	,176*	,029
Adjusted R2	,042	,06	,007	,058	,079	-,016
$\Delta R2$	,003	,06	,005	,046	,133	,013
Sample size	100	89	105	53	39	93

* p < .05.

** p < .01.

Table 4 shows that extraversion (which shows no significant relation when testing H1a) has a significant and positive effect on ambidexterity when conscientiousness is low (-1SD); $\beta = .251$, $p < .05$. Furthermore it shows that extraversion has a significant positive effect on ambidexterity when emotional stability is high (+1SD), $\beta = .38$, $p < .05$. The level of openness to experience did not show a significant effect on the relation between extraversion and ambidexterity.

Thus the relation of extraversion on ambidexterity is dependent on the (low) level of conscientiousness. Hypothesis 1x is therefore accepted. Furthermore it shows that extraversion has a significant positive effect on ambidexterity when emotional stability is high (+1SD), $\beta = .38$, $p < .05$ consistent with Hypothesis 1z.

Hypothesis 1y was not significantly substantiated since neither high nor low levels of openness to experience resulted in an effect with a probability below 5% thus the hypothesis is rejected.

4.3. Moderation effects

Lastly the moderating effect of Autonomy was tested through linear regression. To test this through Process (v 3.4.1) the independent variables (the traits) and the moderator were standardized. The interaction of effect was insignificant ($R^2 = .0012$, $p = .472$). This is related to the general effect of autonomy on all traits.

Next the U-shape of autonomy was tested through regression analysis with high (+1 SD), average (0), and low (-1 SD) levels of autonomy (Cohen, J., & Cohen, P., 1983), (Chung-Yan, 2010).

The levels of autonomy were tested through regression analysis with Gender, Age and Education as model 1 and Autonomy as model 2, Ambidexterity being the dependent variable. Both model 1 and 2 were insignificant at low, average and high levels of Autonomy (table 5). This again is related to only the general effect of autonomy in this case as a direct effect on ambidexterity.

Table 5: General moderation effect of autonomy on the relation between all traits and individual ambidexterity

Dependent Variable: Ambidexterity	R	R Square	R Square Change	B	Std. Error	Beta	t
Model 1	0,088a	0,008	0,008				
Gender				0,194	0,447	0,059	0,433
Age				-0,009	0,02	-0,064	-0,469
Education				0,139	0,315	0,057	0,439
Model 2	0,208	0,043	0,035				
Low Autonomy				0,446	0,304	0,194	1,464

a. Predictors: (Constant), Education, Gender, Age

b. Predictors: (Constant), Education, Gender, Age, Autonomy

Statistical significance: * $p < .05$; ** $p < .01$; *** $p < .001$, N=61

Dependent Variable: Ambidexterity	R	R Square	R Square Change	B	Std. Error	Beta	t
Model 1	0,109	0,012	0,012				
Gender				-0,106	0,179	-0,034	-0,592
Age				-0,008	0,008	-0,061	-1,051
Education level				-0,214	0,163	-0,076	-1,318
Model 2	0,152	0,023	0,011				
Average Autonomy				0,326	0,172	0,107	1,894

a. Predictors: (Constant), Education, Gender, Age

b. Predictors: (Constant), Education, Gender, Age, Autonomy

Statistical significance: *p <.05; **p <.01; ***p <.001, N= 309

Dependent Variable: Ambidexterity	R	R Square	R Square Change	B	Std. Error	Beta	t
Model 1	0,072	0,005	0,005				
Gender				0,166	0,461	0,048	0,359
Age				0,007	0,018	0,048	0,36
Education level				0,09	0,409	0,029	0,219
Model 2	0,207	0,043	0,038				
High Autonomy				1,302	0,875	0,196	1,487

a. Predictors: (Constant), Education, Gender, Age

b. Predictors: (Constant), Education, Gender, Age, Autonomy

Statistical significance: *p <.05; **p <.01; ***p <.001, N= 59

Secondly the levels of autonomy were tested on each of the individual trait through a linear regression model using autonomy as a moderator (after filtering high and low levels (Cohen, J., & Cohen, P., 1983)). To test this through Process (v 3.4.1) the independent variables (the traits) and the moderator were standardized. Ambidexterity being the dependent variable. As seen in table 6 none of the relations were significant at high nor low levels of autonomy.

Table 6: Moderation effect of autonomy (high/low) on the relation between individual traits and individual ambidexterity

Regression of traits with Ambidexterity moderated by autonomy								
Predictors:	Extraversion		Conscientiousness		Emotional stability		Openness to experience	
Autonomy:	High	Low	High	Low	High	Low	Low	Low
Model summary								
R	,225	,205	,192	,199	,224	,273	,216	,282
R ²	,05	,042	,037	,04	,05	,075	,047	,079
MSE	3,015	2,654	3,058	2,661	3,017	2,564	3,027	2,551
F	1,009	,863	,727	,812	1,000	1,588	,933	1,693
P	,395	,465	,54	,492	,399	,202	,431	,178
Model								
B	-,464	-,061	-,151	,301	-1,092	,265	,779	-,874
SE	1,164	,535	,878	,676	1,467	,439	1,27	,641
T	-,398	-,114	-,172	,445	-,744	,604	,613	-1,363
P	,692	,91	,864	,658	,46	,548	,542	,178
P autonomy	,195	,149	,156	,268	,287	,16	,104	,096
P interaction	,815	,907	,888	,644	,415	,996	,493	,345
Sample size	61	63	61	63	61	63	61	63

Autonomy; high is ≥ 1 sd, low ≤ -1 sd

* p < .05.

Hypothesis 2a predicted that the relation of extraversion with ambidexterity was negatively moderated by higher rather than low levels of autonomy. This was correct in the sense that high autonomy resulted in a lower negative B coefficient than in low levels of autonomy but regrettably both were statistically nonsignificant ($>.05$). Therefore Hypothesis 2a is rejected.

The same goes for Hypothesis 2d where openness to experience is positively related to ambidexterity by a high level of autonomy but again nonsignificant. Hypothesis 2b and 2c were incorrect by both the direction of the relation and the level of autonomy and were also nonsignificant.

Concluding all the hypotheses about the moderation effect are rejected. Even though there has been evidence of the effect of autonomy on individual ambidexterity (Tempelaar & Rosenkranz, 2019) and personality (Ng et al., 2008) it does not show a relation in this study. This might have something to do with the contradictory effects of certain traits (e.g. high extraversion versus low conscientiousness for example) and the way they in turn effect the level of ambidexterity through either exploration or exploitation.

5. Discussion

5.1. Theoretical and practical implications

To start with expanding on the research paper of Barrick (2005); Yes, personality matters... in relation to individual ambidexterity. In line with the relatively more comprehensive research (Mao & Washida, 2019) this study similarly found a significant relationship between traits and ambidexterity. This study has shown that some traits are predictors of ambidexterity but in the sense of hints and by no means strong indicators. Even though there is a significant relation between the traits conscientiousness, emotional stability, openness to experiences and ambidexterity they merely explain 7% of the variance together. So 93% is explained by other factors.

This is however one trait (emotional stability) more than Mao et al (2019) found and moreover this study was based on higher quality and validated constructs (Keller & Weibler, 2014) regarding the measurement of ambidexterity. Plus the sample size (435) was more than double compared to Mao et al. (2019).

So how does this work from a managerial perspective? To put it more practically, when acquiring personnel with the goal to increase individual ambidexterity, one might look for higher levels of the latter traits. Big five personality tests are common in selection procedures nowadays. Nevertheless, this would require additional research further exploring the effect of an integrated perspective on personality and ambidexterity.

A first step was taken in this direction by looking at the dependence of extraversion and the significant traits. The relationship between extraversion and ambidexterity was non-significant across the entire studied population, but I did find that relatively unconscientious and emotional stable persons were significantly more ambidextrous than others.

To use a metaphor, personality is like a Rubik's cube, all facets are influenced by each other and one must take the whole cube into account not just one side. Thus this might be a direction for future research.

This being said one has to take the work context into account as well. In this study the level of autonomy was accounted for but showed no significant effect. Neither high or low levels of autonomy as a moderator, on individual traits nor all traits, resulted in a significant relation with ambidexterity. This seems to contradict the described effect of reduced autonomy (Bidmon et al, 2019; Tempelaar et al, 2019). Although the former research was based upon observations of strategy workshops and this study uses a self-assessment of autonomy. This might be an explanation of the contrast in outcomes. Another clarification might be sought in the direction of the contradictory effects autonomy has on certain traits and the way they in turn effect the level of ambidexterity through either exploration or exploitation.

5.2. Limitations

Firstly, this study involves cross-sectional data, a specific situation at a certain time is reflected in the survey data. Additionally a limitation related with the timeframe of collecting this data is that respondents reflect on an average working week whilst being in lockdown due to Covid-19. A longitudinal study would extend this research by providing insights based on change and development regarding personality and ambidexterity and contextual factors.

Second, an additional limitation is individual respondent bias, they needed to reflect on their own behaviour and personality. Are their responses reflecting who they are or who they want to be? Part of this bias reflects on the next limitation.

Third, as mentioned in chapter 3 regarding the reliability of the TIPI, even though it fulfilled its purpose of a short measurement, it remains a weaker link in this study. As mentioned in regard to the low reliability of the TIPI acquiescence bias could be a factor as well as individual respondent bias. Future studies will benefit when they extend the measurement of the personality test. Both in reliability and in-depth perspective on personality in connection to ambidexterity.

Fourth, the measurement of ambidexterity has two approaches (the combined or balanced dimension (Cao et al., 2009)) in ambidexterity literature. This study was conducted using the balanced dimension. It would be sensible to create or decide on a validated construct of individual ambidexterity to further develop this line of research. This is crucial if scholars want to further examine all the facets of an individual in relation to ambidexterity. Of course there might not be one true form of individual ambidexterity but creating distinction between organisational and individual measurements might be beneficial.

Fifth, the moderating effect of autonomy was not analysed on hypotheses regarding dependence of traits. When expanding this research on the combination of all traits in a profile it will be interesting to control for autonomy also.

Sixth, even though the sample size was very sound it could have been more heterogenic regarding generalization purposes. In the sense that all respondents were teachers of vocational education. A broader cross section of different industries would be preferable. This was accounted for by an identical survey distributed externally but due to low response rate it was ignored. A last limitation and notion in relation to generalisation purposes is that this study is based on a WEIRD sample. WEIRD in the meaning of Western, Educated, Industrialized, Rich and Democratic, which according to a study of top psychology journals in 2008, found that in the vast majority of researches 96% of subjects were from Western industrialized countries which represents just 12% of the world's population (Henrich, Heine & Norenzayan, 2010).

5.3. Future research

In extension of the third limitation, future research could expand the personality component (more items per trait) of this research to strengthen reliability of the traits. And use more facets per trait (Costa & McCrae, 1995) to further examine underlying factors influencing individual ambidexterity.

Additionally, expand on the interdependence of traits in relation to ambidexterity. As mentioned before using a complete personality profile might shed light on certain ratio's between traits that cumulate to a high ambidextrous profile.

As ambidexterity is a balanced dimension (in this study), so might the “optimal” personality profile related to ambidexterity be. If this is true one can extend on understanding ambidexterity by adding contextual predictors (i.e. autonomy, leadership (perhaps also in line with personality), task design, remuneration, organisational context or external dynamism).

A final avenue for future research might be a reverse analysis. When looking at the highest levels of ambidexterity in this research and building a personality profile. Then validating if this profile indeed has a predictive value in relation to ambidexterity. Or perhaps the profile is an expression of a underlying ratio of traits, who knows. The golden ratio of ambidexterity is still undiscovered.

5.4. Conclusion

This study and its supporting empirical data revealed that certain personality traits (and some interdependencies) are predictors of individual ambidexterity. This conclusion should be an incentive to further unravel the effect of complete personality profiles on individual ambidexterity. This study contributes to research on individual ambidexterity by encouraging a personality or identity perspective.

5.5. Acknowledgements

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7. Appendix

7.1. Survey

Beste collega,

Dank je voor het invullen van deze enquête. Zoals ik in mijn e-mail aangaf onderzoek ik hoe organisaties wendbaarder kunnen worden. Via deze enquête wil ik onderzoeken of persoonlijkheid een voorspeller kan zijn voor wendbaarheid. Je krijgt vier blokken met stellingen te zien met daarbij enkele aanvullende vragen.

Neem bij het invullen van de enquête een normale werkweek in gedachten.

Alvast hartelijk dank.

Met vriendelijke groet,

Menno Poort

Survey:

Q3

Ik ben een

- ☐ Vrouw
- ☐ Man

Q6

Mijn leeftijd is

Q7

Mijn hoogst genoten opleiding

- ☐ Vmbo, havo-, vwo-onderbouw en mbo1
- ☐ Havo, vwo en mbo2-4
- ☐ Hbo- en wo-bachelor
- ☐ Hbo- en wo-master of doctor

Q1

Voor de persoonlijkheidstest krijg je 10 stellingen.

"Ik zie mijzelf als..."

	Ze oneens	Redelijk oneens	Klein beetje oneens	Niet oneens, niet eens	Klein beetje eens	Redelijk eens	Ze mee eens
Extravert, enthousiast	☐	☐	☐	☐	☐	☐	☐
Kritisch, strijdzuchtig	☐	☐	☐	☐	☐	☐	☐
Betrouwbaar, gedisciplineerd	☐	☐	☐	☐	☐	☐	☐
Angstig, snel overstuurt	☐	☐	☐	☐	☐	☐	☐
Open voor nieuwe ervaringen, complex	☐	☐	☐	☐	☐	☐	☐
Gereserveerd, stil	☐	☐	☐	☐	☐	☐	☐
Sympathiek, warm	☐	☐	☐	☐	☐	☐	☐
Slordig, achteloos	☐	☐	☐	☐	☐	☐	☐
Kalm, emotioneel stabiel	☐	☐	☐	☐	☐	☐	☐
Behoudend, niet creatief	☐	☐	☐	☐	☐	☐	☐

Q8

Hoeveel jaar werk je al in je huidige functie?

Q9

Hoeveel jaar werk je al in je huidige organisatie?

Q10

In welke sector ben je werkzaam (werkveld van je college)?

- | | |
|---|---|
| ☐ Landbouw bosbouw en visserij | ☐ Financiële dienstverlening |
| ☐ Delfstoffenwinning | ☐ Verhuur en handel van onroerend goed |
| ☐ Industrie | ☐ Specialistische zakelijke diensten |
| ☐ Energievoorziening | ☐ Verhuur en overige zakelijke diensten |
| ☐ Waterbedrijven en afvalbeheer | ☐ Openbaar bestuur en overheidsdiensten |
| ☐ Bouwnijverheid | ☐ Onderwijs |
| ☐ Handel | ☐ Gezondheids en welzijnszorg |
| ☐ Vervoer en opslag | ☐ Cultuur sport en recreatie |
| ☐ Horeca | ☐ Overige dienstverlening |
| ☐ Informatie en communicatie | |

Page Break

Q4

Voor het meten van wendbaarheid krijg je twee keer 8 stellingen.

"Ik houd mij wekelijks bezig met enerzijds...

	Ze er oneens	Rede lijk oneens	Klein beetje oneens	Niet oneens, niet eens	Klein beetje eens	Rede lijk eens	Ze er mee eens
Activiteiten waar ik moet omgaan met voor mij onbekende situaties	☐	☐	☐	☐	☐	☐	☐
Activiteiten die zo complex zijn dat ze in het begin moeilijk te overzien zijn.	☐	☐	☐	☐	☐	☐	☐
Activiteiten waarin ik voorheen onbekend terrein betreed.	☐	☐	☐	☐	☐	☐	☐
Activiteiten die veel aanpassingsvermogen van mijn kant vereist.	☐	☐	☐	☐	☐	☐	☐
Activiteiten die een compleet andere strategie vereisen.	☐	☐	☐	☐	☐	☐	☐
Activiteiten waarbij ik niet de competenties heb die nodig zijn om ze uit te voeren totdat ik ze daadwerkelijk uitvoer.	☐	☐	☐	☐	☐	☐	☐
Activiteiten waarvan de gevolgen op het moment van uitvoering nog niet precies te voorzien zijn.	☐	☐	☐	☐	☐	☐	☐
Activiteiten waarbij ik de grenzen van mijn kennis bereik.	☐	☐	☐	☐	☐	☐	☐

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Q5

"Ik houd mij wekelijks bezig met anderzijds..."

	Ze oneens	Redelijk oneens	Klein beetje oneens	Niet oneens, niet eens	Klein beetje eens	Redelijk eens	Ze mee eens
Activiteiten die ik heel routinematig uitvoer.	☐	☐	☐	☐	☐	☐	☐
Activiteiten die ik uitvoer volgens een bekend patroon.	☐	☐	☐	☐	☐	☐	☐
Activiteiten waarop ik goed ben voorbereid.	☐	☐	☐	☐	☐	☐	☐
Activiteiten waarvan de uitvoering volledig duidelijk is.	☐	☐	☐	☐	☐	☐	☐
Veel voorkomende activiteiten.	☐	☐	☐	☐	☐	☐	☐
Gemakkelijk te plannen activiteiten.	☐	☐	☐	☐	☐	☐	☐
Activiteiten die binnen een vooraf gedefinieerde periode kunnen worden uitgevoerd.	☐	☐	☐	☐	☐	☐	☐
Activiteiten die verwijzen naar een duidelijk omschreven probleemgebied.	☐	☐	☐	☐	☐	☐	☐

Page Break

Q11

Met onderstaande 9 stellingen onderzoek ik de invloed van vrijheid in je werkzaamheden op je wendbaarheid.

Hoeveel vrijheid heb je in je werk?

	Ze oneens	Redelij oneens	Klein beetje oneens	Niet oneens, niet eens	Klein beetje eens	Redelij eens	Ze mee eens
Ik ben vrij om te beslissen hoe ik mijn werk af wil krijgen	☐	☐	☐	☐	☐	☐	☐
Ik ben vrij om te kiezen hoe ik mijn werk moet uitvoeren	☐	☐	☐	☐	☐	☐	☐
Ik kan in mijn team kiezen hoe ik wil werken	☐	☐	☐	☐	☐	☐	☐
Ik kan beslissen wanneer ik bepaalde activiteiten in mijn team moet uitvoeren	☐	☐	☐	☐	☐	☐	☐
Ik heb controle over de planning van mijn werk in mijn team	☐	☐	☐	☐	☐	☐	☐
Ik heb enige controle over de volgorde van mijn activiteiten in mijn team	☐	☐	☐	☐	☐	☐	☐
Ik kan zelf bepalen wat mijn doelstellingen zijn	☐	☐	☐	☐	☐	☐	☐
Ik heb enige controle over wat ik in mijn team moet bereiken	☐	☐	☐	☐	☐	☐	☐
Ik kan de manier waarop ik word geëvalueerd beïnvloeden, dus ik kan sommige aspecten van wat ik doe benadrukken en andere bagatelliseren	☐	☐	☐	☐	☐	☐	☐

Q12

In mijn functiebeschrijving staan activiteiten die zich vooral kenmerken als

- ☐ Routinematige en veel voorkomende activiteiten
- ☐ Improviserende en vernieuwende activiteiten
- ☐ Beide bovenstaande activiteiten

Q13

In mijn perceptie vraagt mijn functie om activiteiten die zich vooral kenmerken als

- ☐ Routinematige en veel voorkomende activiteiten
- ☐ Improviserende en vernieuwende activiteiten
- ☐ Beide bovenstaande activiteiten

Q14

In verband met de Corona maatregelen kan ik mij voorstellen dat dit door o.a. thuiswerken invloed heeft gehad op je werk. De huidige periode vraagt om activiteiten die zich vooral kenmerken als

- ☐ Routinematige en veel voorkomende activiteiten
- ☐ Improviserende en vernieuwende activiteiten
- ☐ Beide bovenstaande activiteiten

Q17

Hartelijk dank voor het invullen.

Fijne dag!

7.2. Survey items factor loading

Table 7: Factor loadings of the constructs Exploration, Exploitation and Autonomy

Item	Loading
Exploration (Keller and Weibler, 2015, $\alpha=.84$)	
Activities in which you have to deal with previously unknown situations.	,700
Activities that are so complex that they are difficult to survey at the start.	,756
Activities in which you enter previously unknown territory.	,770
Activities that require a good deal of adaptability on your part.	,711
Activities that require a completely different strategy.	,754
Activities in which you do not acquire the competences required for carrying them out until you actually carry them out.	,601
Activities whose consequences are not yet exactly foreseeable at the time they are carried out.	,686
Activities in which you reach the limits of your knowledge.	,563
Exploitation (Keller and Weibler, 2015, $\alpha=.84$)	
Activities that you carry out very routinely.	,659
Activities you carry out in accordance with a familiar pattern.	,742
Activities for which you are well prepared.	,701
Activities whose execution is completely clear.	,743
Frequently recurring activities.	,714
Easily plannable activities.	,721
Activities that can be carried out within a previously defined period.	,664
Activities that refer to a clearly defined problem area.	,600
Autonomy (Langfred, 2005, $\alpha=.85$)	
I am free to decide how to go about getting my work done	,706
I am free to choose how to carry out my work	,721
I am able to choose the way to go about my work in the team	,740
I can decide when to do particular activities as part of my work in the team	,763
I have control over the scheduling of my work in the team	,745
I have some control over the sequencing of my activities in the team	,679
I am able to decide for myself what my objectives are	,671
I have some control over what I am supposed to accomplish in the team	,685
I can influence how I am evaluated, so I can emphasize some aspects of what I do and play down others	,483

Note: All items are on a 1-to-7 scale, n=435

